

SIMATS ENGINEERING

ASSESSMENT TOOL 1: Class Test

COURSE NAME: Cloud Computing and
Big Data Analytics

COURSE CODE: CSA1522

COURSE OUTCOME COVERED

CO4: *Analyze big data characteristics and challenges across domains including security and application aspects.*
(BL4)

Dave's Taxonomy: Articulation (Level 4)
Question Count: 10

Total Marks: 30

Code of Conduct

I Divyasree.M/192524098 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student :*Divyasree.M*

Assessment Tasks

Class Test

1. Analyze a multinational retail organization struggling with fragmented customer data across online and offline channels, and propose a unified big data architecture that enables real-time customer analytics and personalized recommendations.
2. Evaluate the effectiveness of distributed computing models in processing large-scale scientific research data, and justify how parallel processing improves computational efficiency and scalability compared to centralized systems.
3. Design a fraud detection framework for a digital payment platform using big data analytics, and justify how streaming data processing and predictive models improve fraud prevention accuracy.
4. Assess the challenges of storing and processing multimedia data such as video, audio, and images in a big data environment, and recommend suitable technologies and storage mechanisms for efficient management.
5. Construct a disaster recovery and business continuity strategy for a cloud-based big data platform, and justify how redundancy, replication, and automated failover mechanisms ensure operational resilience.
6. Analyze the impact of data quality issues such as inconsistency, duplication, and missing values in a large-scale analytics platform, and propose data governance strategies to improve analytical reliability.
7. Design a scalable recommendation system for an online streaming service using big data technologies, and justify how user behavior analytics and distributed processing enhance recommendation accuracy and user engagement.
8. Evaluate the role of cloud computing in modern big data ecosystems, and critically examine its advantages and limitations in terms of scalability, cost management, flexibility, and security.
9. Develop a real-time analytics framework for transportation and logistics management, and justify how big data technologies improve route optimization, fuel efficiency, and operational decision-making.

10. Critically analyze ethical and legal challenges associated with big data collection and analytics, and recommend organizational policies that ensure responsible data usage, regulatory compliance, and user trust.

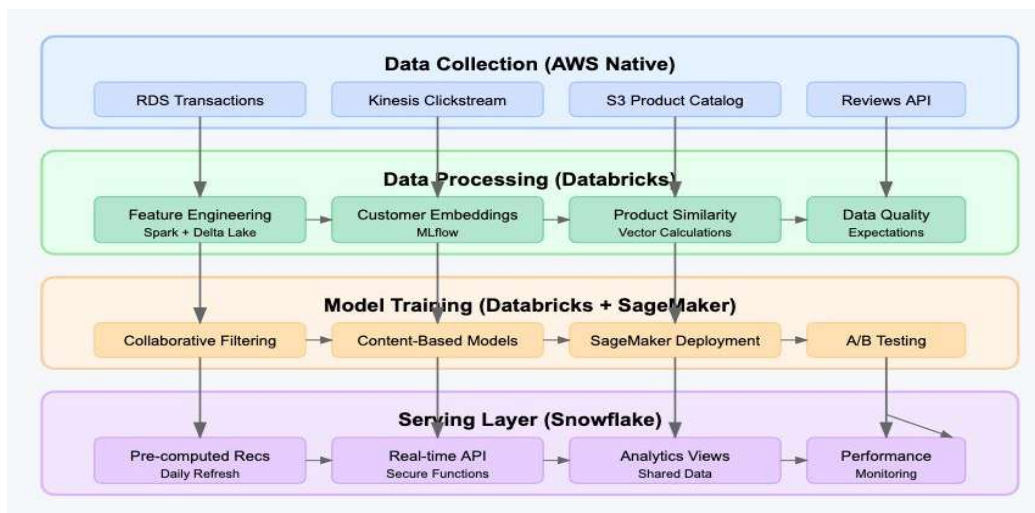
Class Test Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Concept Accuracy	30%	Accurate and well-expressed technical answers	Mostly accurate answers	Partial accuracy	Inaccurate answers
Coverage	20%	Covers all required points	Covers most points	Limited coverage	Very poor coverage
Use of Examples	15%	Uses relevant and meaningful examples	Some relevant examples	Weak examples	No useful examples
Analytical Awareness	20%	Shows strong awareness of issues and implications	Reasonable awareness	Basic awareness	Minimal awareness
Clarity	15%	Clear and concise writing	Generally clear	Some unclear expressions	Poor clarity

1. Analyze a multinational retail organization struggling with fragmented customer data across online and offline channels, and propose a unified big data architecture that enables real-time customer analytics and personalized recommendations.

1. Unified Big Data Architecture for Retail:

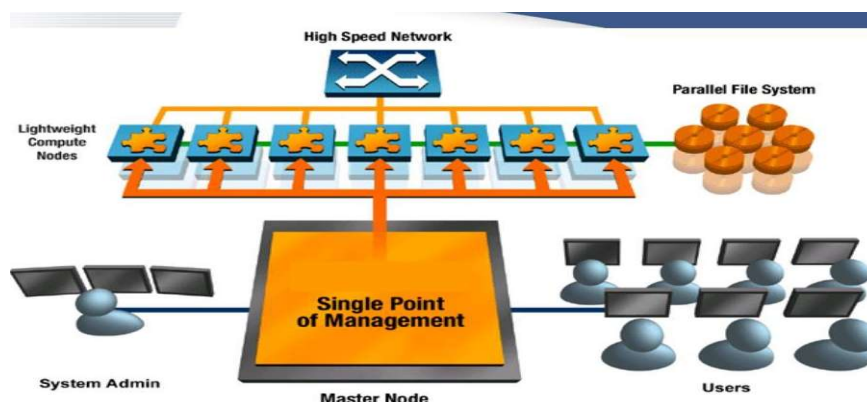
A multinational retailer should use a unified big data architecture that combines online and offline customer data into a single platform. Data from e-commerce websites, mobile apps, POS systems, and loyalty programs can be collected using Apache Kafka, stored in a Data Lake (HDFS or Amazon S3), and processed by Apache Spark for real-time analytics. Machine learning models then generate personalized product recommendations. For example, a customer who buys shoes in a physical store can instantly receive matching accessory suggestions on the mobile app. This architecture improves customer experience, enables targeted marketing, and provides a complete 360-degree customer view.



2. Evaluate the effectiveness of distributed computing models in processing large-scale scientific research data, and justify how parallel processing improves computational efficiency and scalability compared to centralized systems.

Distributed Computing in Scientific Research

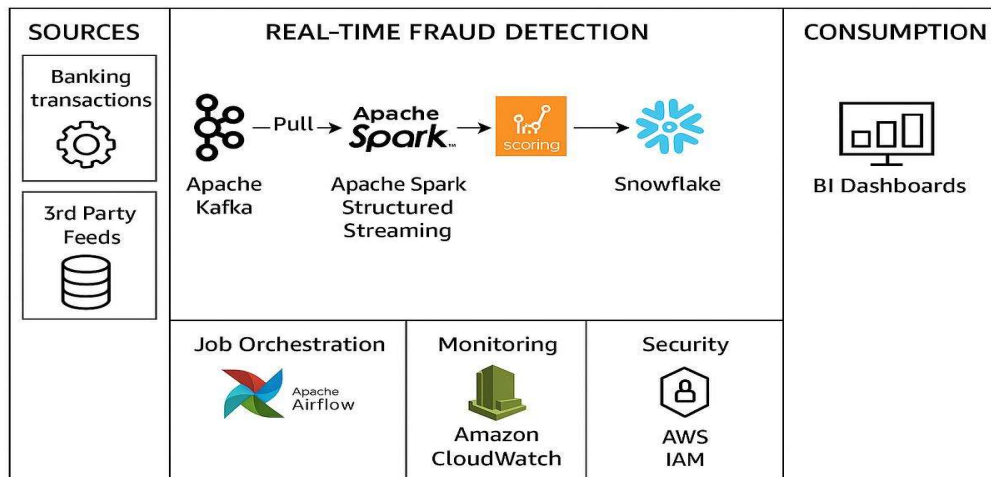
Distributed computing processes large scientific datasets by dividing work among multiple computers instead of relying on a single system. Technologies like Hadoop and Apache Spark allow parallel processing, where several nodes analyze different parts of a dataset simultaneously. For example, CERN uses distributed computing to process massive particle physics data from the Large Hadron Collider. Compared to centralized systems, distributed computing offers faster processing, better scalability, and fault tolerance, making it suitable for research fields such as genomics, astronomy, and climate modeling.



3. Design a fraud detection framework for a digital payment platform using big data analytics, and justify how streaming data processing and predictive models improve fraud prevention accuracy.

Fraud Detection for Digital Payments

A fraud detection framework for a digital payment platform should combine streaming analytics with predictive machine learning models. Transaction data is collected through Apache Kafka, analyzed instantly using Spark Streaming, and evaluated by algorithms such as Random Forest or anomaly detection. The system checks unusual transaction amounts, locations, and device usage in real time. For example, if a customer suddenly makes a purchase from another country, the transaction can be flagged for verification. This approach reduces fraud losses by identifying suspicious activities before transactions are completed.



4. Assess the challenges of storing and processing multimedia data such as video, audio, and images in a big data environment, and recommend suitable technologies and storage mechanisms for efficient management.

Managing Multimedia Data in Big Data

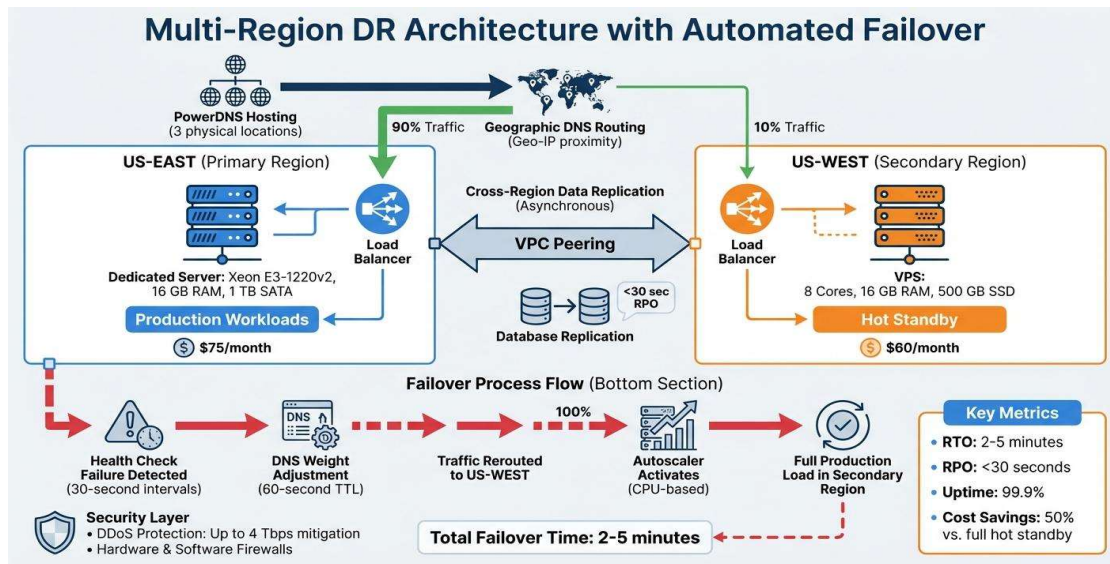
Storing videos, images, and audio creates challenges because multimedia files are large, arrive continuously, and exist in different formats. Cloud object storage such as Amazon S3, HDFS, and Elasticsearch for metadata management help organize and retrieve these files efficiently. Compression techniques like H.265 reduce storage requirements, while distributed processing improves performance. For example, Netflix stores and streams millions of videos using cloud-based storage and analytics. These technologies provide scalability, faster access, and efficient multimedia management.



5. . Construct a disaster recovery and business continuity strategy for a cloud-based big data platform, and justify how redundancy, replication, and automated failover mechanisms ensure operational resilience. Disaster

Recovery for Cloud Big Data Platforms

A cloud-based big data platform should use redundancy, replication, and automated failover to ensure business continuity. Data should be replicated across multiple regions, while backup servers remain available if one system fails. Automated failover through cloud services or Kubernetes allows applications to continue running without manual intervention. For example, Amazon S3's cross-region replication protects data during regional outages. This strategy minimizes downtime, prevents data loss, and ensures reliable operations during disasters.



6. Analyze the impact of data quality issues such as inconsistency, duplication, and missing values in a large-scale analytics platform, and propose data governance strategies to improve analytical reliability.

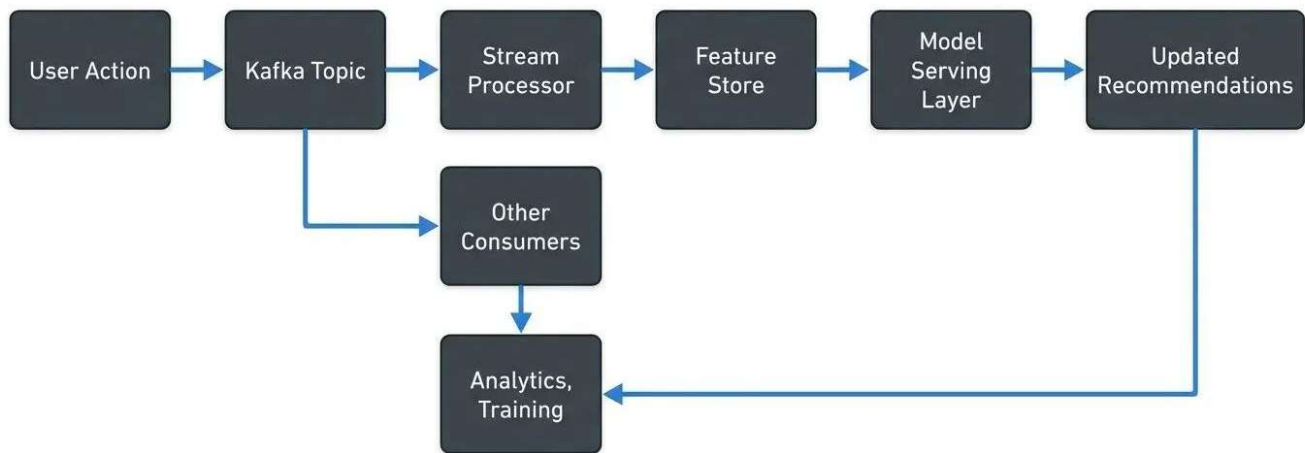
Data Quality and Governance

Data quality issues such as inconsistency, duplicate records, and missing values reduce the accuracy of analytics. Organizations should implement data governance through validation rules, Master Data Management (MDM), metadata management, and automated data cleansing using Apache Spark. For example, duplicate customer records can be merged into a single profile for accurate reporting. Good data governance improves analytical reliability, supports better decision-making, and helps organizations comply with data regulations.

7. Design a scalable recommendation system for an online streaming service using big data technologies, and justify how user behavior analytics and distributed processing enhance recommendation accuracy and user engagement.

Recommendation System for Streaming Services

A scalable recommendation system analyzes user behavior such as watch history, ratings, and search patterns to suggest relevant content. Apache Kafka collects user activity, while Apache Spark processes large datasets across multiple servers. Machine learning techniques like collaborative filtering improve recommendation accuracy. For example, Netflix recommends similar movies immediately after a user finishes watching a film. Distributed processing enables faster recommendations and increases user engagement by delivering personalized content.



8. Evaluate the role of cloud computing in modern big data ecosystems, and critically examine its advantages and limitations in terms of scalability, cost management, flexibility, and security.

Role of Cloud Computing in Big Data

Cloud computing plays a major role in big data by providing scalable storage and computing resources on demand. Services like AWS, Microsoft Azure, and Google Cloud reduce infrastructure costs through pay-as-you-go pricing while supporting global deployment. However, organizations must manage challenges such as vendor lock-in, data privacy, and security risks. For example, Spotify uses cloud infrastructure to process billions of streaming events. Overall, cloud computing offers flexibility and scalability but requires proper security and cost management.

9. Develop a real-time analytics framework for transportation and logistics management, and justify how big data technologies improve route optimization, fuel efficiency, and operational decision-making.

Real-Time Analytics for Transportation

A real-time analytics framework uses GPS, IoT sensors, and traffic data to improve transportation and logistics operations. Apache Kafka collects live data, while Spark Streaming analyzes vehicle location, fuel usage, and road conditions instantly. For example, UPS optimizes delivery routes using real-time traffic information. This approach reduces fuel consumption, shortens delivery times, and helps managers make faster operational decisions through live dashboards.

10. Critically analyze ethical and legal challenges associated with big data collection and analytics, and recommend organizational policies that ensure responsible data usage, regulatory compliance, and user trust.

Ethical and Legal Challenges in Big Data

Big data collection raises ethical issues such as privacy violations, algorithmic bias, and lack of transparency. Organizations must also comply with laws like GDPR and India's DPDP Act. They should implement policies for user consent, data encryption, access control, and regular security audits. For example, companies can provide privacy dashboards that allow users to manage their personal data. Responsible data governance builds user trust while ensuring legal compliance and ethical use of information.

